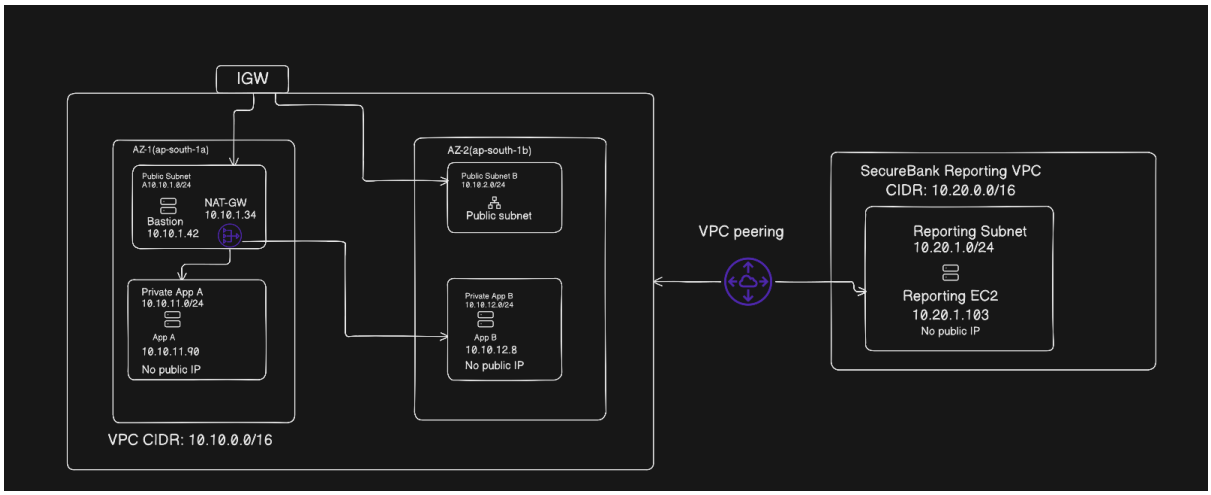
SecureBank AWS VPC Project: Final Submission

This document covers the six required deliverables for the SecureBank multi-tier VPC lab: a production VPC (10.10.0.0/16) peered with a private reporting VPC (10.20.0.0/16).

1: VPC architecture diagram with CIDRs and Availability Zones.



2: subnet and route-table mapping.

Subnet Mapping:

The SecureBank network infrastructure is divided into a Production VPC and a Reporting VPC. The Production VPC uses the CIDR block 10.10.0.0/16, while the Reporting VPC uses 10.20.0.0/16.

The Production VPC contains public and private subnets distributed across two Availability Zones to improve availability and support future growth. Public subnets host the Bastion Host and NAT Gateway, while private subnets host the application servers without direct public internet access.

VPC	Subnet	CIDR	Purpose
Production	Public A	10.10.1.0/24	Bastion and NAT Gateway

Production	Public B	10.10.2.0/24	Public subnet for future growth
Production	Private App A	10.10.11.0/24	Private application server
Production	Private App B	10.10.12.0/24	Private application server
Reporting	Reporting subnet	10.20.1.0/24	Private reporting server

Route-Table Mapping:

Route tables determine how traffic moves between subnets, the internet, and the connected Reporting VPC. The public route table directs internet-bound traffic to the Internet Gateway. The private route table uses the NAT Gateway for outbound internet access and VPC Peering for private communication with the Reporting VPC.

Route table	**Destination**	**Target**
Production Public RT	10.10.0.0/16	Local
Production Public RT	0.0.0.0/0	Internet Gateway
Production Private RT	10.10.0.0/16	Local
Production Private RT	0.0.0.0/0	NAT Gateway
Production Private RT	10.20.0.0/16	VPC Peering
Reporting RT	10.20.0.0/16	Local
Reporting RT	10.10.0.0/16	VPC Peering

3: Security-group and NACL rule tables:

Security groups and Network Access Control Lists (NACLs) are used to protect the SecureBank AWS network. Security groups control inbound and outbound traffic at the EC2 instance level, while NACLs control traffic at the subnet level.

In this project, security groups and NACLs are configured to restrict unauthorized access, allow required communication between application servers, and support secure connectivity between the Production VPC and the Reporting VPC.

Bastion Security Group

The Bastion Security Group controls access to the Bastion Host located in the public subnet. SSH access is restricted to the administrator's public IP address.

Direction	Protocol	Port	Source/Destination	Action
Inbound	TCP	22	Administrator's public IP /32	Allow

Application Servers Security Group

The Application Security Group protects App A and App B in the private subnets. SSH access is permitted from the Bastion Security Group, preventing direct SSH access from the public internet.

Direction	Protocol	Port	Source/Destination	Action
Inbound	TCP	22	Bastion Security Group	Allow

Reporting Security Group

The Reporting Security Group controls access to the Reporting EC2 instance. HTTP traffic on port 8080 is allowed from the two private application subnets.

Direction	Protocol	Port	Source/Destination	Action
Inbound	TCP	8080	10.10.11.0/24 — App A subnet	Allow
Inbound	TCP	8080	10.10.12.0/24 — App B subnet	Allow

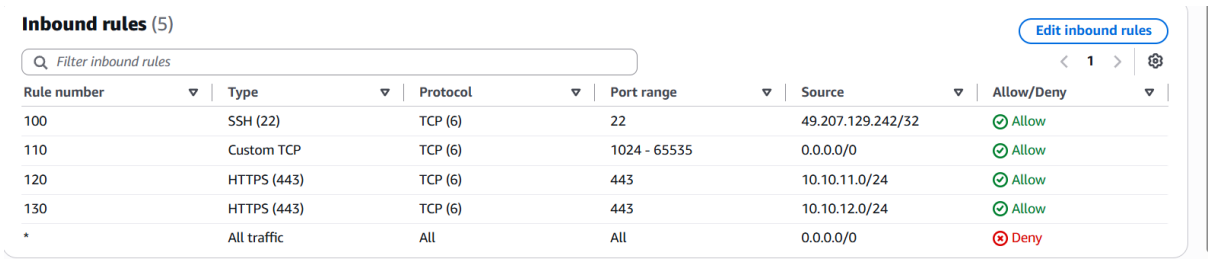
Network Access Control Lists (NACLs)

Network Access Control Lists (NACLs) provide an additional layer of security for the SecureBank VPC infrastructure. They control inbound and outbound traffic at the subnet level. NACLs are stateless, meaning that both incoming requests and their return traffic must be permitted by the appropriate rules.

In this project, custom NACLs are configured for the public subnets, private application subnets, and Reporting subnet. These rules restrict unnecessary traffic while allowing the communication required for administration, outbound internet access, and private communication between the Production and Reporting VPCs.

Public NACL

The Public NACL protects the public subnets containing the Bastion Host and NAT Gateway. It permits the required administrative SSH traffic and traffic needed for internet connectivity.



Outbound rules (8)

Edit outbound rules

Filter outbound rules

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
100	SSH (22)	TCP (6)	22	10.10.11.0/24	Allow
110	SSH (22)	TCP (6)	22	10.10.12.0/24	Allow
120	HTTPS (443)	TCP (6)	443	0.0.0.0/0	Allow
130	HTTP (80)	TCP (6)	80	0.0.0.0/0	Allow
140	Custom TCP	TCP (6)	1024 - 65535	0.0.0.0/0	Allow
150	Custom TCP	TCP (6)	1024 - 65535	10.10.11.0/24	Allow
160	Custom TCP	TCP (6)	1024 - 65535	10.10.12.0/24	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Private NACL

The Private NACL protects the private application subnets hosting App A and App B. It permits required traffic involving the Bastion subnet, outbound HTTPS connections through the NAT Gateway, and communication with the Reporting VPC.

Inbound rules (3)

Edit inbound rules

Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	SSH (22)	TCP (6)	22	10.10.1.0/24	Allow
110	Custom TCP	TCP (6)	1024 - 65535	10.20.0.0/16	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Outbound rules (5)

Edit outbound rules

Filter outbound rules

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
13	SSH (22)	TCP (6)	22	10.20.0.0/16	Allow
100	Custom TCP	TCP (6)	1024 - 65535	10.10.1.0/24	Allow
110	HTTPS (443)	TCP (6)	443	0.0.0.0/0	Allow
120	Custom TCP	TCP (6)	1024 - 65535	10.20.0.0/16	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Reporting NACL

The Reporting NACL protects the Reporting subnet containing the Reporting EC2 instance. It allows HTTP traffic on port 8080 from the App A and App B subnets. It also permits the required ephemeral TCP ports for return traffic to the application servers.

Inbound rules (5)

Edit inbound rules

Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	HTTP* (8080)	TCP (6)	8080	10.10.11.0/24	Allow
110	HTTP* (8080)	TCP (6)	8080	10.10.12.0/24	Allow
120	SSH (22)	TCP (6)	22	10.10.11.0/24	Allow
130	SSH (22)	TCP (6)	22	10.10.12.0/24	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Outbound rules (5)

Edit outbound rules

Filter outbound rules

1

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
100	Custom TCP	TCP (6)	1024 - 65535	10.10.11.0/24	Allow
110	Custom TCP	TCP (6)	1024 - 65535	10.10.12.0/24	Allow
120	SSH (22)	TCP (6)	22	10.10.11.0/24	Allow
130	SSH (22)	TCP (6)	22	10.10.12.0/24	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

4: Screenshots showing bastion access, private outbound access, and successful VPC peering

Bastion Host Access

The Bastion Host is deployed in the public subnet and provides a controlled administrative access point to the private application servers. SSH access is restricted through security group rules.

```
PS C:\Users\Abdulsalam> cd .\Downloads\
PS C:\Users\Abdulsalam\Downloads> ssh -i "SecureBank.pem" ec2-user@ec2-3-110-30-175.ap-south-1.compute.amazonaws.com
The authenticity of host 'ec2-3-110-30-175.ap-south-1.compute.amazonaws.com (3.110.30.175)' can't be established.
ED25519 key fingerprint is SHA256:hUjfO5dw7DP5MyqhePBG2j6QDAWbZGDizLBj7QXsNMQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-110-30-175.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-10-1-42 ~]$ ls
[ec2-user@ip-10-10-1-42 ~]$ ll
```

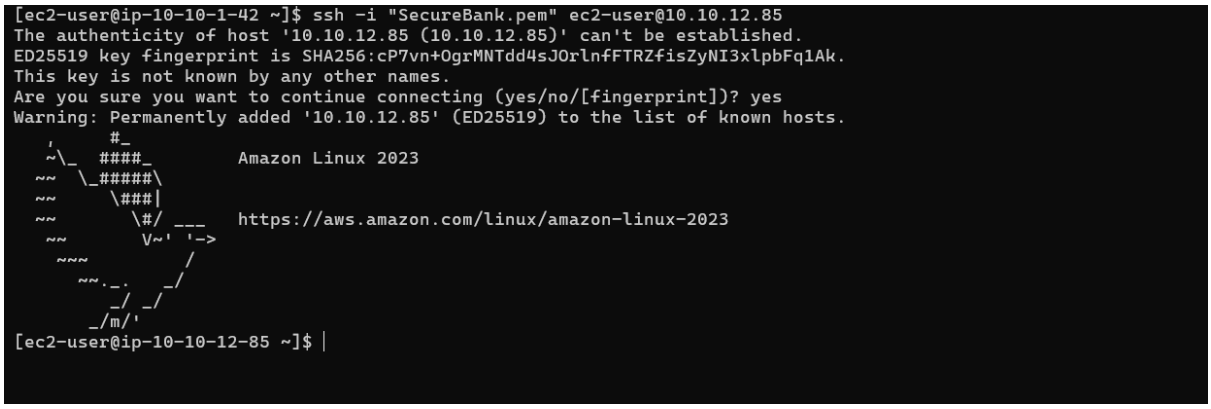
Figure: Bastion SSH access

Private Outbound Internet Connectivity

App A and App B are deployed in private subnets without public IPv4 addresses. The NAT Gateway provides outbound internet connectivity while the private instances remain inaccessible through direct public IP connections.

```
[ec2-user@ip-10-10-1-42 ~]$ ssh -i "SecureBank.pem" ec2-user@10.10.11.90
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-10-11-90 ~]$
[ec2-user@ip-10-10-11-90 ~]$
```

Figure: Private outbound internet access



VPC Peering Connectivity

The Production VPC and Reporting VPC are connected through an active VPC Peering connection. Routes in both VPCs direct traffic destined for the other VPC through the peering connection.

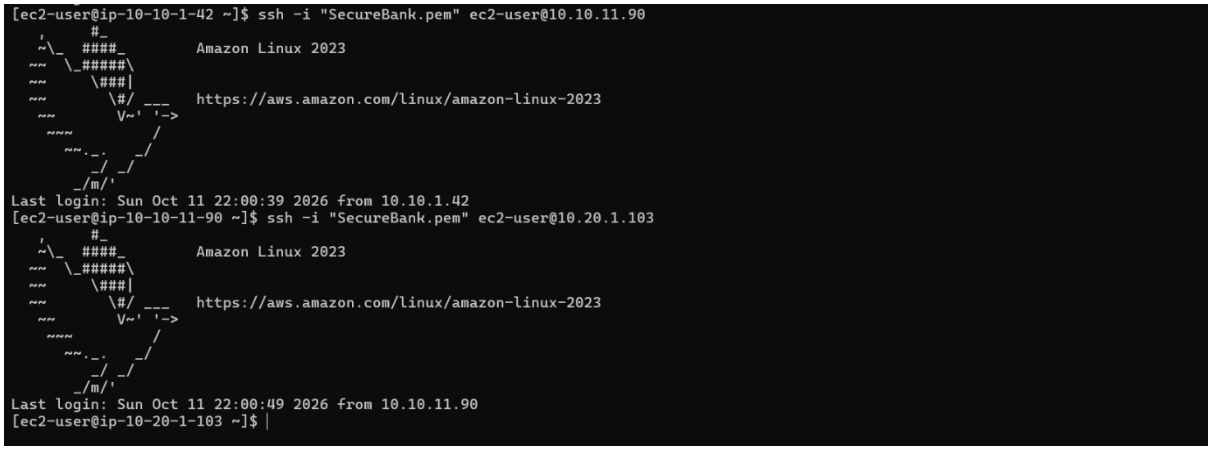


Figure: VPC peering connectivity

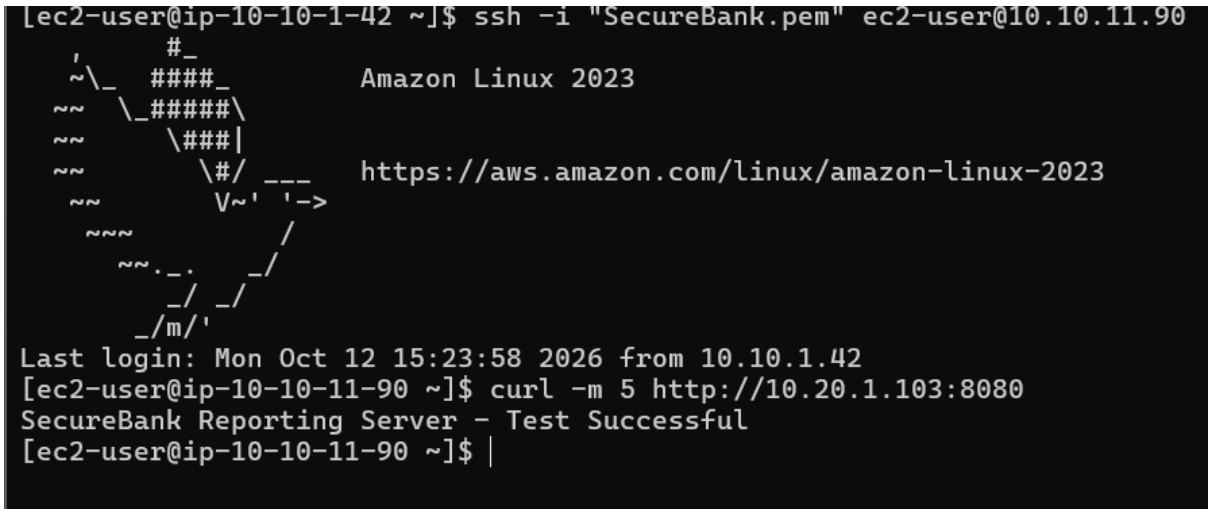
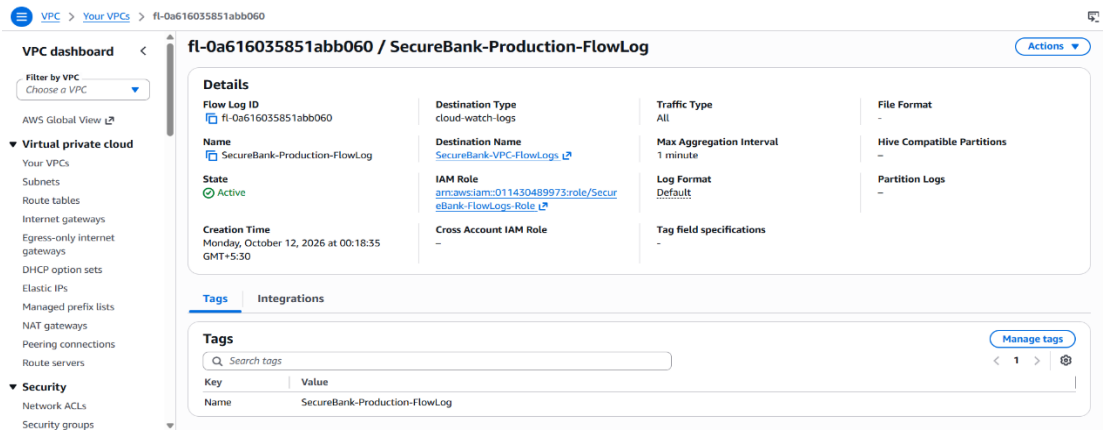


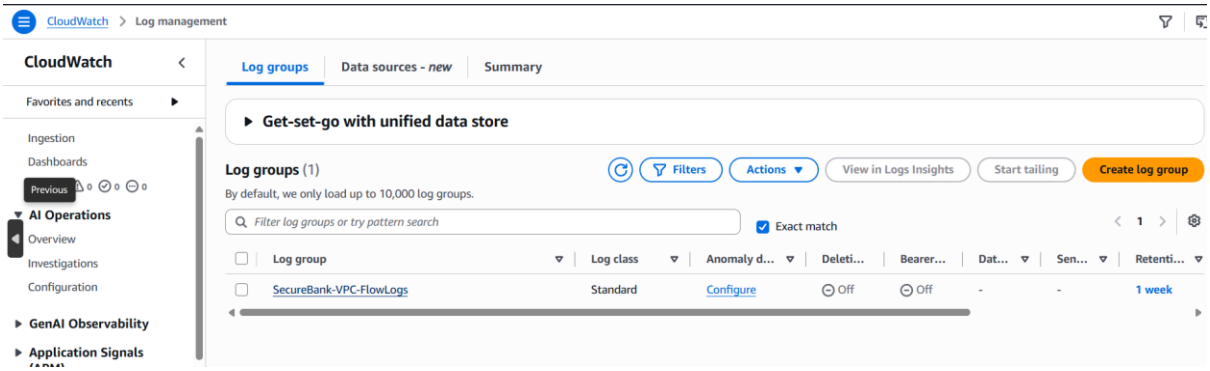
Figure: Successful VPC Peering Connectivity

5: Flow Log evidence for accepted and rejected traffic.

VPC Flow Logs were configured to monitor network traffic within the SecureBank AWS environment. The logs were delivered to the CloudWatch log group SecureBank-VPC-FlowLogs.

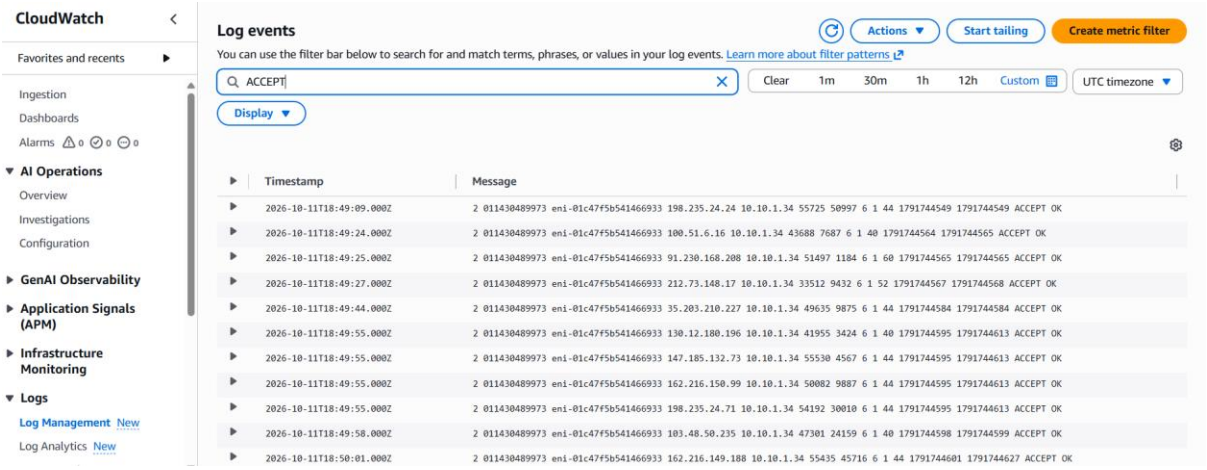






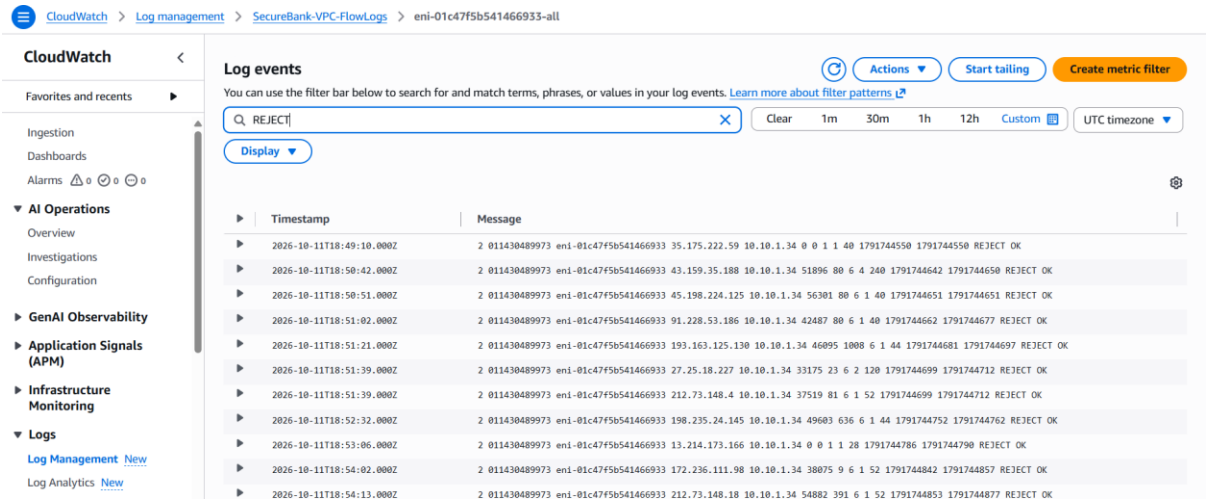
Accepted Traffic

The Flow Logs contain records marked ACCEPT, indicating that the logged network traffic was permitted by the applicable network controls.



Rejected Traffic

The Flow Logs also contain records marked REJECT, indicating that the logged traffic was rejected by the applicable network controls.



6: Incident Report

Incident Title

Reporting VPC Connectivity Failure on Port 9090

Problem Description

During connectivity testing, an HTTP request from App A in the Production VPC to the Reporting EC2 instance failed when accessing

port 9090. The request timed out, indicating that communication on this port was unsuccessful.

Diagnosis

The network configuration was reviewed, including the Reporting Security Group and Reporting Network ACL. Port 8080 was permitted for communication from the application subnets, while port 9090 was not permitted by the configured network rules.

Resolution

The application was tested on port 8080, which was already permitted by the network security configuration. The Reporting HTTP server was configured to return a simple test message from a dedicated directory.

Verification

The connectivity tests produced the following results:

- **Port 9090:** The request timed out, confirming that communication on this port was unsuccessful.
- **Port 8080:** The request returned SecureBank Reporting Server - Test Successful, confirming successful communication.

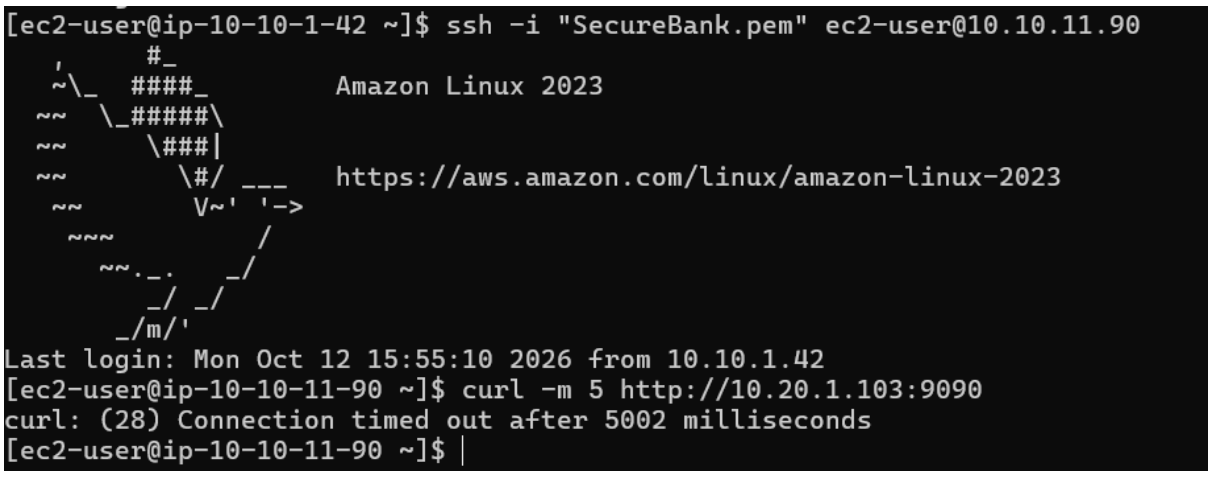


Figure: Failed connectivity test on port 9090.

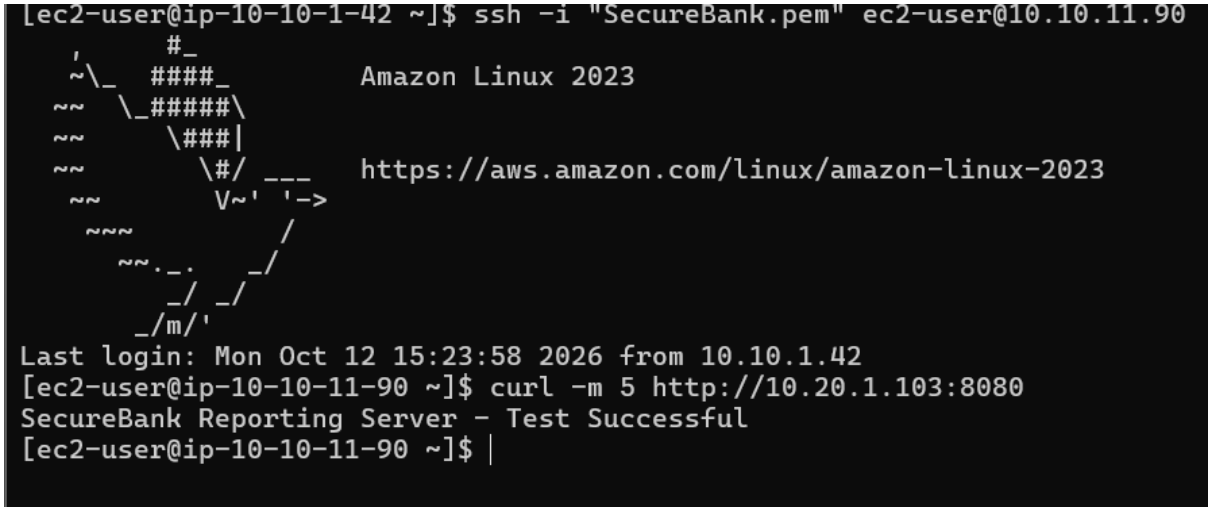


Figure : Successful connectivity test on port 8080.